DRIVEABILITY—IDLE VIBRATION—VEHICLES EQUIPPED WITH 2.0L ZETEC ENGINE AND AUTOMATIC TRANSAXLE ONLY—SERVICE TIPS

Article No. 02-21-12

FORD: 2000-2002 ESCORT ZX2

ISSUE

Some vehicles equipped with a 2.0L Zetec engine and automatic transaxle may exhibit a vibration felt at idle. This may be caused by the engine mounting system.

ACTION

Diagnose and service idle vibration by referring to the following Idle Vibration Service Tips.

SERVICE INFORMATION

NOTE

ENGINE MOUNT #3 BRACKET REPLACEMENT DIRECTIVE APPLIES ONLY TO VEHICLES BUILT PRIOR TO 04/10/2002. VEHICLES PRODUCED AFTER THIS DATE ALREADY HAVE THE UPDATED/IMPROVED ENGINE MOUNT #3 BRACKET INSTALLED FROM THE FACTORY. ON VEHICLES BUILT AFTER 04/10/2002, PERFORM ONLY THE ENGINE SUPPORT CROSSMEMBER (BELLY BAND) PROCEDURE.

NOTE

GRILL OPEN REINFORCEMENT (GOR) LOWER ATTACHING BOLTS DIRECTIVE APPLIES ONLY TO VEHICLES BUILT PRIOR TO 8/28/2001. VEHICLES PRODUCED AFTER THIS DATE ALREADY HAVE THE INSULATOR BOLT INSTALLED FROM THE FACTORY.

NOTE

THE PROCEDURE CONTAINED IN THIS TSB ARTICLE MAY NOT COMPLETELY ELIMINATE ALL VIBRATION FROM THE VEHICLE. SOME VIBRATION IS INHERENT IN I-4 ENGINE DESIGNS, SUCH AS ZETEC 2.0L ENGINE. THIS PROCEDURE IS DESIGNED TO REDUCE THE AMOUNT OF VIBRATION TO AN ACCEPTABLE LEVEL..

NOTE

IF INCORRECTLY POSITIONED, THE ENGINE MOUNTS CAN TRANSMIT EXCESSIVE ENGINE VIBRATION TO THE BODY, RESULTING IN UNACCEPTABLE IDLE PERFORMANCE. THE SEQUENCE OF THE INDICATED STEPS IS CRITICAL AND MUST BE PERFORMED IN THE ORDER INDICATED.

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IDLE VIBRATION TIPS

ENGINE MOUNT #3 BRACKET REPLACEMENT PROCEDURE, VEHICLES BUILT PRIOR TO 04/10/2002

1. Be sure engine is warm.
2. Shift the transaxle into Neutral and apply the parking brake.
3. Remove the engine support crossmembers using the following procedure:
   a. Install electrical system memory saver.
   b. Disconnect the positive battery cable.
   c. Support engine with Three-Bar Engine Support Tool (Rotunda 303-D063) and Lifting Brackets (Rotunda 303-050).
   d. Raise Vehicle. Refer to appropriate model year Workshop Manual, Section 100-02.
   e. Remove the rear lower crossmember (attached with four (4) bolts (Figure 1)).
   f. Loosen the center bolt on the #1 (rear) lower engine mount (Figure 3).
   g. Loosen the center bolt on the #2 (front) lower engine mount (Figure 3).
   h. Remove the engine support crossmember that runs from front to rear of the engine (Belly Band). This can be accomplished by removing the front two (2) attaching bolts, the rear two (2) attaching nuts and the four (4) engine mount #1 and #2 nuts (Figure 1).

NOTE
DISCARD THE TWO ENGINE MOUNT #1 FLAT WASHERS ONLY FOR VEHICLES BUILT PRIOR TO 12/03/2001. VEHICLES PRODUCED AFTER THIS DATE ALREADY HAVE THE FLAT WASHERS REMOVED FROM THE FACTORY.

   i. Modify the Belly Band on the two rear attaching engine mount #1 holes by slotting holes approximately 1/4″ (6 mm). Use a Drill Bit (Figure 2).
   j. Reinstall the front attaching points of Belly Band with original two (2) Bolts. Apply 2-3 drops of Threadlocker (EOAZ-19554-AA) to the bolts (Figure 1).
   k. Reinstall rear-attaching points of Belly Band with original two (2) nuts (Figure 1).
   l. Torque the retaining nuts and bolts to 40-58 Lb-ft (54-76 N•m).
   m. Install the engine mount #1 two (2) nuts (rear) (Figure 4).
   n. Install the engine mount #2 two (2) nuts (front) (Figure 4).
   o. Torque the retaining nuts to 26-38 Lb-ft (38-52 N•m).

NOTE
VEHICLES PRODUCED AFTER 06/25/2001 ALREADY HAVE THE NEW SPACERS INSTALLED FROM THE FACTORY.

4. Reinstall the rear lower crossmember and install two (2) new spacer (2S4Z-5L020-AA) (Figure 5).
5. Torque the retaining nuts and bolts to 40-58 Lb-ft (54-76 N•m).
6. Lower the vehicle.
7. Remove the #3 engine mount and damper (Figure 6).

NOTE
PAINT THE NEW BRACKET (2S4Z-6028-AA) TO MATCH WITH THE BODY EXTERIOR COLOR.

8. Remove the bracket engine mount #3 and replace with new bracket (2S4Z-6028-AA) (Figure 7).
9. Torque the retaining bolts to 12-17 Lb-ft (16-22 N•m).
10. Reinstall the #3 engine mount. Leave the center through-bolt loose.
11. Remove the Three-Bar Engine Support Tool and lifting brackets.
12. Restores battery connections.
13. Start the engine with the vehicle on the ground. Launch the vehicle both forward and reverse in a normal manner. Allow the engine to assume a neutral position in the mounts, turn off the engine. This allows the engine to establish a neutral position.
14. Torque the center through-bolt of the #3 engine mount to 50-68 Lb-ft (67-93 N•m).

15. Lift the vehicle.
16. Torque the #2 (front) engine mount center bolt to 50-68 Lb-ft (67-93 N•m).
17. Torque the #1 (rear) engine mount center bolt to 50-68 Lb-ft (67-93 N•m).
18. After complete this procedure, disconnect the vehicle battery (5 minutes, minimum) to reset KAM, or on newer vehicles, use the "KAM Reset" feature on the New Generation Star (NGS) Tester or Worldwide Diagnostic System (WDS) Tester.

ENGINE SUPPORT CROSSMEMBER (BELLY BAND) PROCEDURE

1. Warm the engine to operating temperature.
2. Shift the transaxle into Neutral and apply the parking brake.
3. Remove the engine support crossmembers using the following procedure:
   a. Install electrical system memory saver.
   b. Disconnect the positive battery cable.
   c. Support engine with Three-Bar Engine Support Tool (Rotunda 303-D063) and Lifting Brackets (Rotunda 303-050).
   d. Raise Vehicle. Refer to appropriate model year Workshop Manual, Section 100-02.
   e. Remove the rear lower crossmember (attached with four (4) bolts (Figure 1)).
   f. Remove the engine support crossmember that runs from front to rear of the engine (Belly Band). This can be accomplished by removing the front two (2) attaching bolts, the rear two (2) attaching nuts and the four (4) engine mount #1 and #2 nuts (Figure 1).
   g. Modify the Belly Band on the two rear attaching engine mount #1 holes by slotting holes approximately 1/4″ (6 mm). Use a Drill Bit (Figure 2).
   h. Loosen the center bolt on the #1 (rear) lower engine mount (Figure 3).
   i. Loosen the center bolt on the #2 (front) lower engine mount (Figure 3).

j. Reinstall the front attaching points of Belly Band with original two (2) Bolts. Apply 2-3 drops of Theadlocker (EOAZ-19554-AA) to the bolts (Figure 1).

k. Reinstall rear-attaching points of Belly Band with original two (2) nuts (Figure 1).

l. Torque the retaining nuts and bolts to 40-58 Lb-ft (54-76 N•m).

m. Install the engine mount #1 two (2) nuts (rear) (Figure 4).

n. Install the engine mount #2 two (2) nuts (front) (Figure 4).

o. Torque the retaining nuts to 26-38 Lb-ft (38-52 N•m).

4. Reinstall the rear lower crossmember.

5. Torque the retaining nuts and bolts to 40-58 Lb-ft (54-76 N•m).

6. Lower the vehicle.

7. Loosen the center through-bolt #3 engine mount and damper (Figure 6).

8. Remove the Three-Bar Engine Support Tool and lifting brackets.

9. Restore battery connections.

10. Start the engine with the vehicle on the ground. Launch the vehicle both forward and reverse in a normal manner. Allow the engine to assume a neutral position in the mounts. Turn off the engine. This allows the engine to establish a neutral position.

11. Torque the center through-bolt of the #3 engine mount to 50-68 Lb-ft (67-93 N•m).

12. Lift the vehicle.

13. Torque the #2 (front) engine mount center bolt to 50-68 Lb-ft (67-93 N•m).

14. Torque the #1 (rear) engine mount center bolt to 50-68 Lb-ft (67-93 N•m).

15. After you complete this procedure, disconnect the vehicle battery (5 minutes minimum) to reset KAM, or on newer vehicles, use the "KAM Reset" feature on the New Generation Star (NGS) Tester or Worldwide Diagnostic System (WDS) Tester.

GRILL OPEN REINFORCEMENT (GOR) LOWER ATTACHING BOLTS PROCEDURES, VEHICLES BUILT PRIOR TO 8/28/2001

1. Remove the front bumper cover (Fascia). Refer to the appropriate model year Workshop Manual, Section 501-19.

2. Install the grill open reinforcement lower attaching bolts and nut spacer (W520203-S nut (2) and W701389-S309 bolt (2)). Torque the bolt to 46.9-64.6 Lb-in (5.3-7.3 N•m). Refer to appropriate model year Workshop Manual, Section 501-02. (Figure 8).

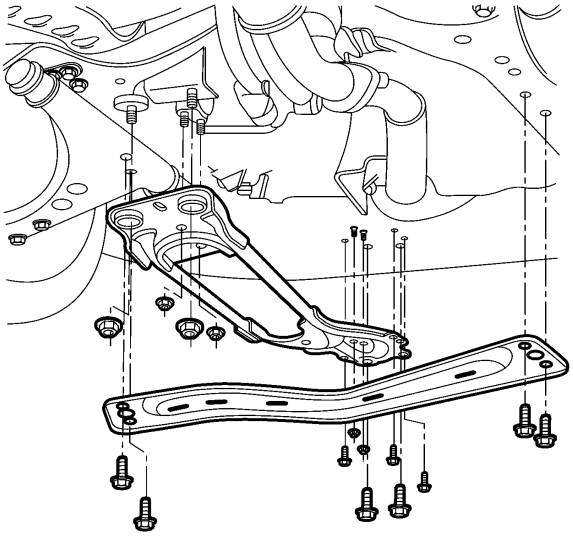
3. Reinstall the front bumper cover (Fascia). Refer to the appropriate model year Workshop Manual, Section 501-19.

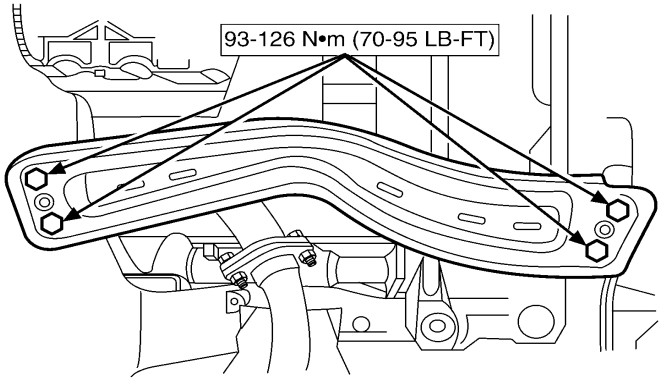
ADDITIONAL NOISE/VIBRATION DIAGNOSTIC TIPS

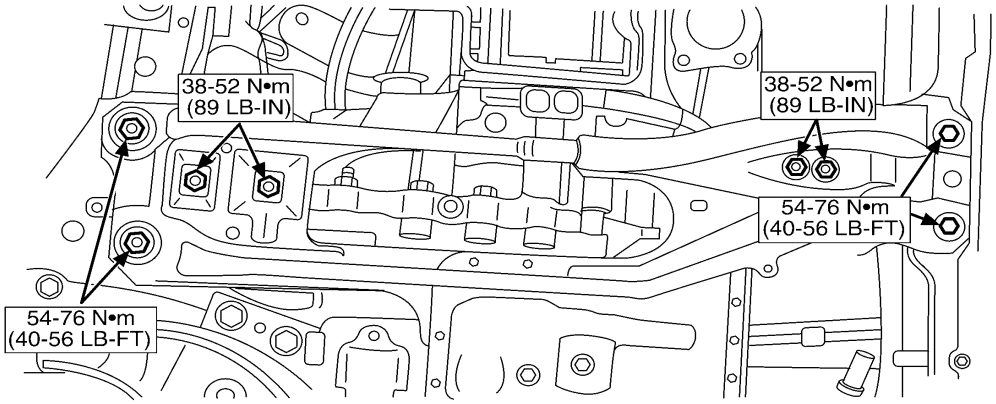
- **Fuel Tank Flange:** The leading edge flange of the tank could contact the tank support strap. This contact, though not presenting a risk of damage to the tank or strap, could result in a noise from the floor of the vehicle. This issue can be resolved by adding a small rubber edge protector to the flange at the tank straps. Follow the appropriate Escort Workshop Manual, Section 310-01, and make sure the flange protector is positioned correctly (Figure 9).
- **Front Fascia:** With the help of an assistant, determine if the front fascia is vibrating. The assistant should place their hands on the fascia while the engine is running in gear (be sure parking brake is applied). If a change in vibration is noticed in the vehicle, loosen the top three (3) fascia retaining bolts, pull the fascia upward, and tighten. Vehicles that have a front license plate are more prone to vibrate at idle than fascias without plates (To eliminate noise from front or rear license plates it is also recommended to add rubber insulation behind plates).
- **Hood Snubbers and Seals:** If the concern is related to hood shake, confirm contact of the rubber snubbers at the front corners of the hood. Confirm that the intermediate snubbers attached to the fender assembly are contacting the hood. If they do not contact the hood, lower the adjustable snubbers at the front corners until the intermediate snubbers contact the hood. If this adjustment causes a poor margin appearance between the fenders and the hood, shim the intermediate snubber with a small piece of foam wedged between the intermediate snubber and the fender. Verify the rubber seal at the cowl side of the hood is installed completely.
- **Exhaust System:** Be sure exhaust system hangers are not over compressed or distorted. After performing engine mount procedures, loosen the intermediate exhaust flange in front of the flexible connection. Allow the system to neutralize and retorque the system. Verify that all exhaust system heat shields are tight and edges are not contacting floor pan sheet metal (Figure 10).
- **Clean Air Tube:** Confirm that the clean air tube is correctly assembled to the throttle body. Confirm that the assembled components between the air cleaner assembly and the throttle body are completely assembled and all affected clamps are tightened. Air leakage at these connections will degrade idle stability and result in increased idle Noise/Vibration/Harshness (NVH) (Figure 11).

PART NUMBER	PART NAME
2S4Z-5L020-AA	Spacer
2S4Z-6028-AA	Bracket - Engine Mount # 3
W520203-S	Nut
W701389-S309	Bolt
EOAZ-19554-AA	Threadlock And Sealer

OTHER APPLICABLE ARTICLES: 00-5-9
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 608000, 608400, 703000, 703100







TB-6804-A

Figure 1 - Article 02-21-12

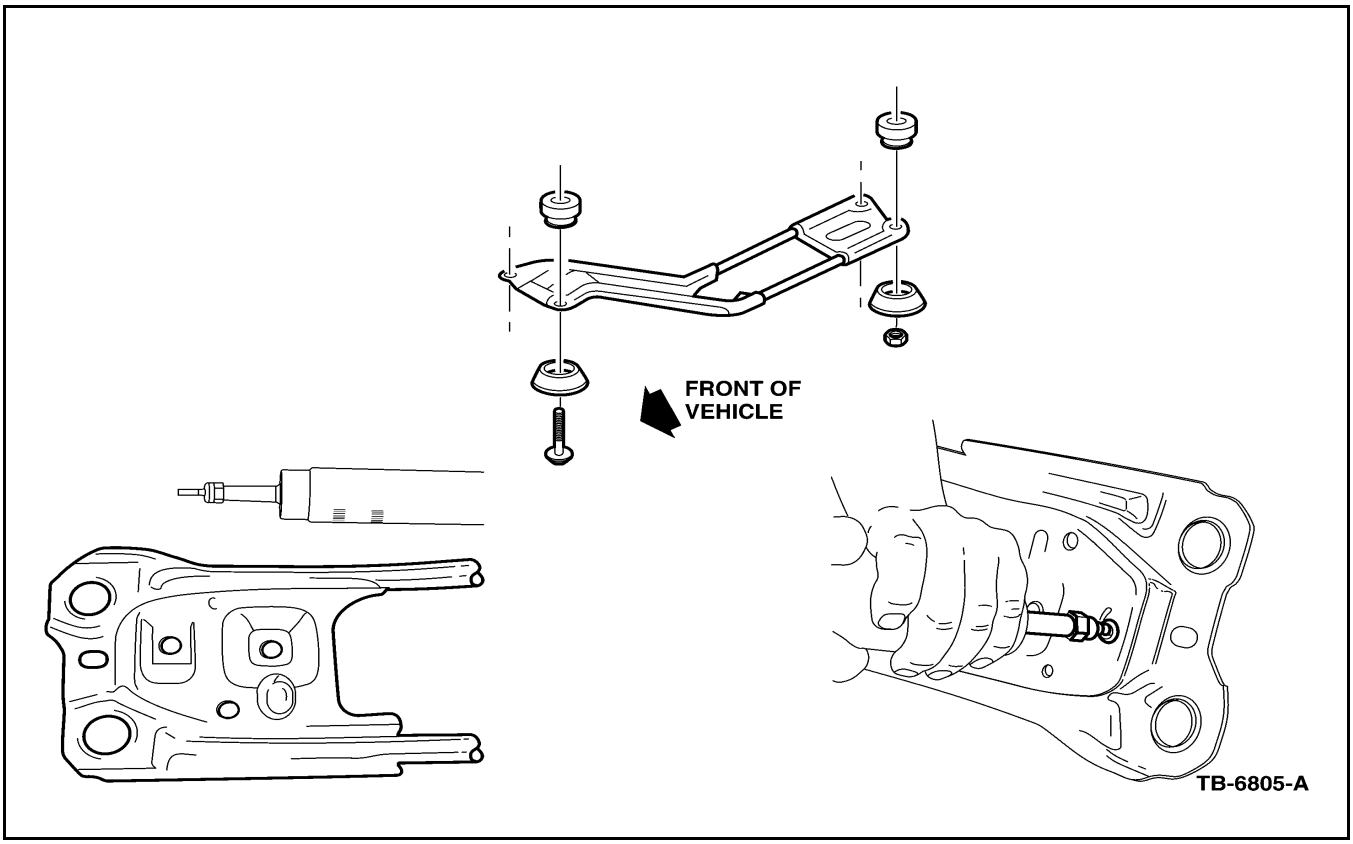


Figure 2 - Article 02-21-12

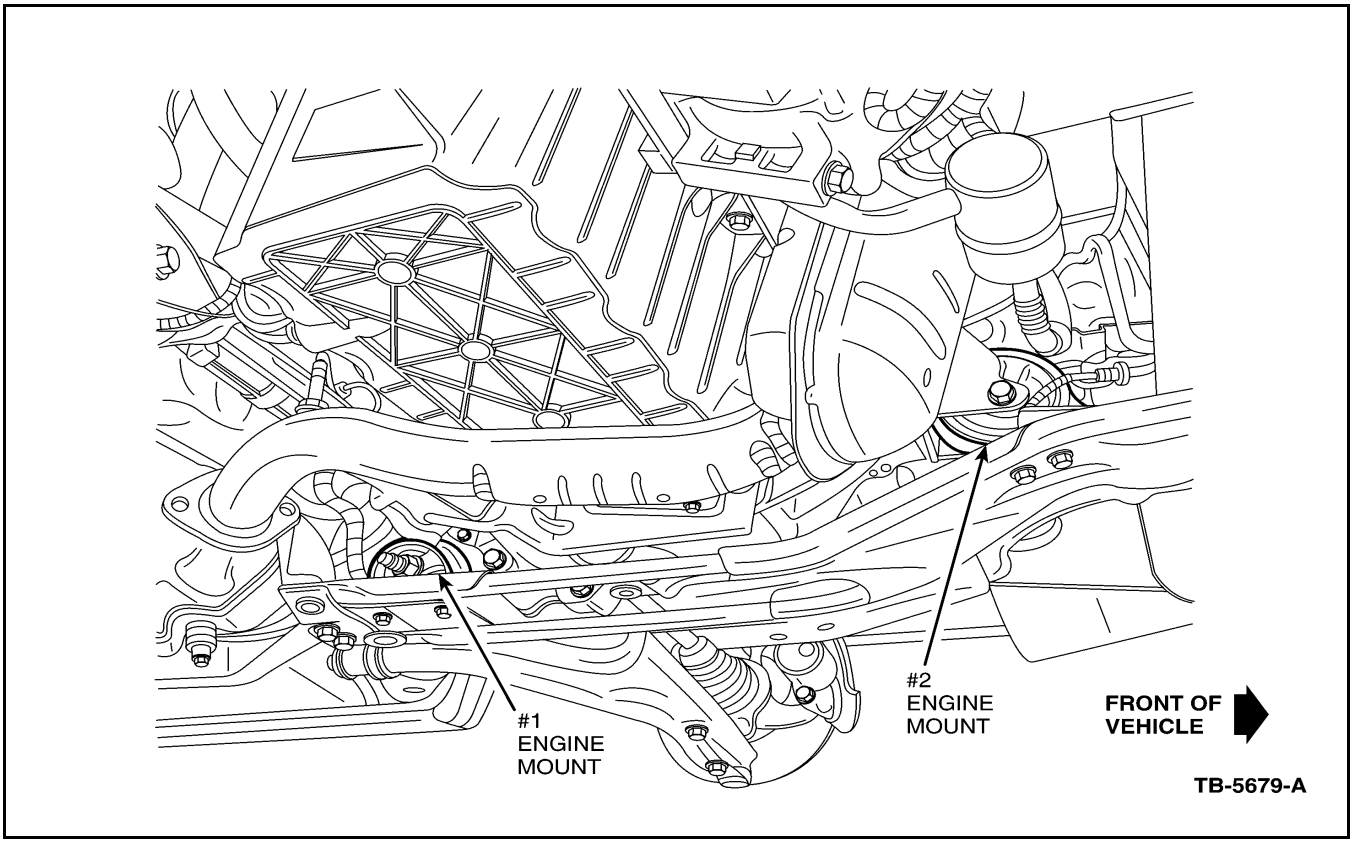


Figure 3 - Article 02-21-12

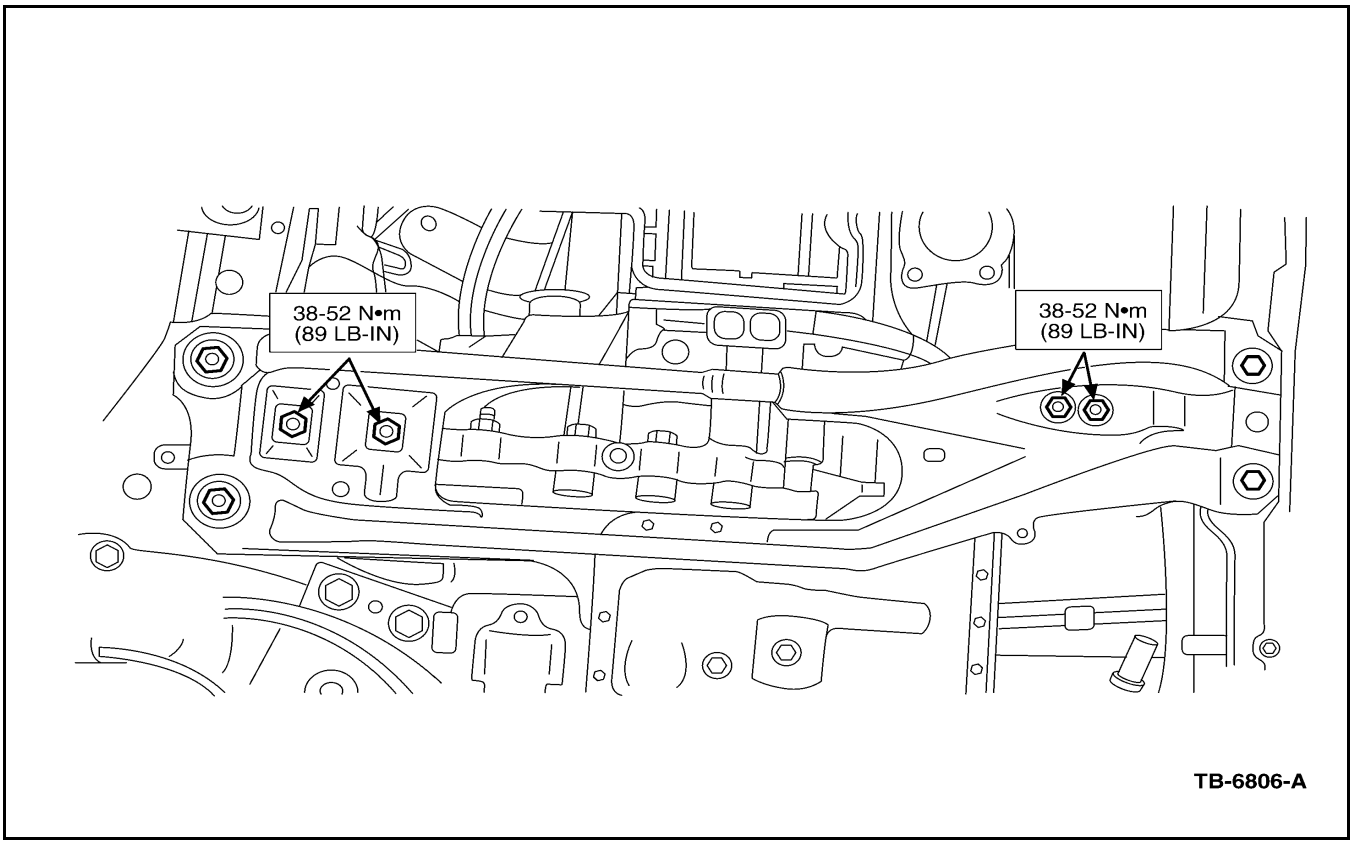


Figure 4 - Article 02-21-12

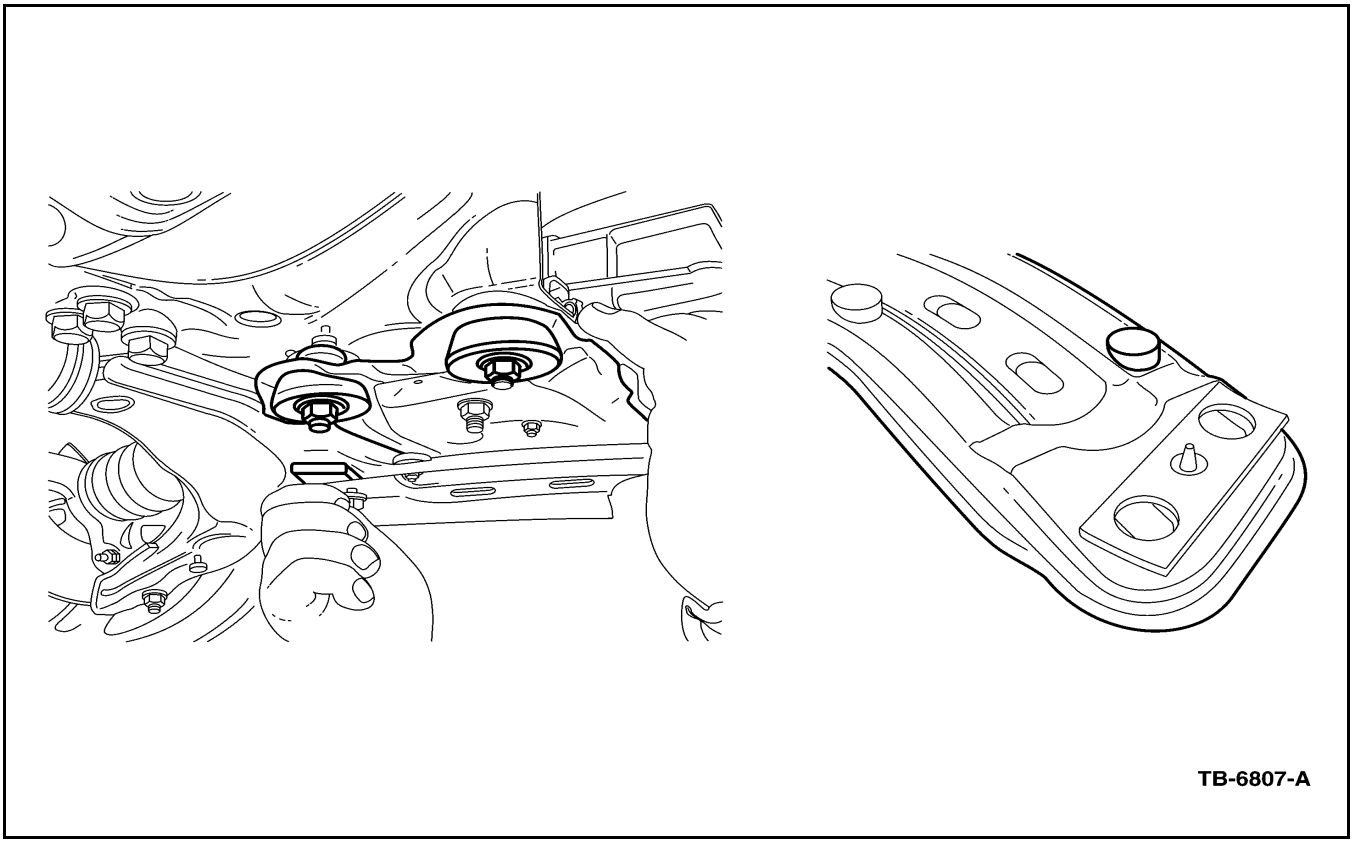


Figure 5 - Article 02-21-12

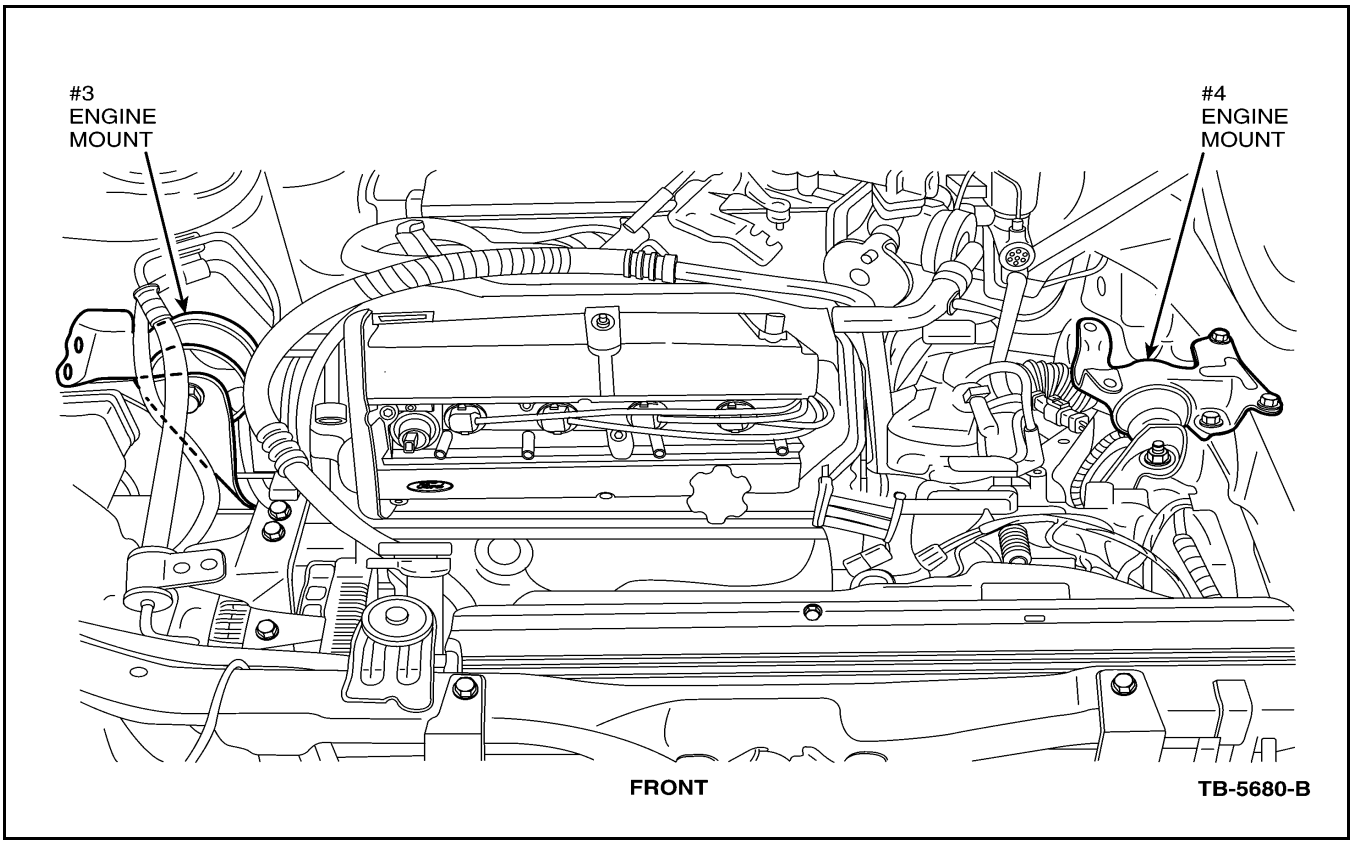


Figure 6 - Article 02-21-12

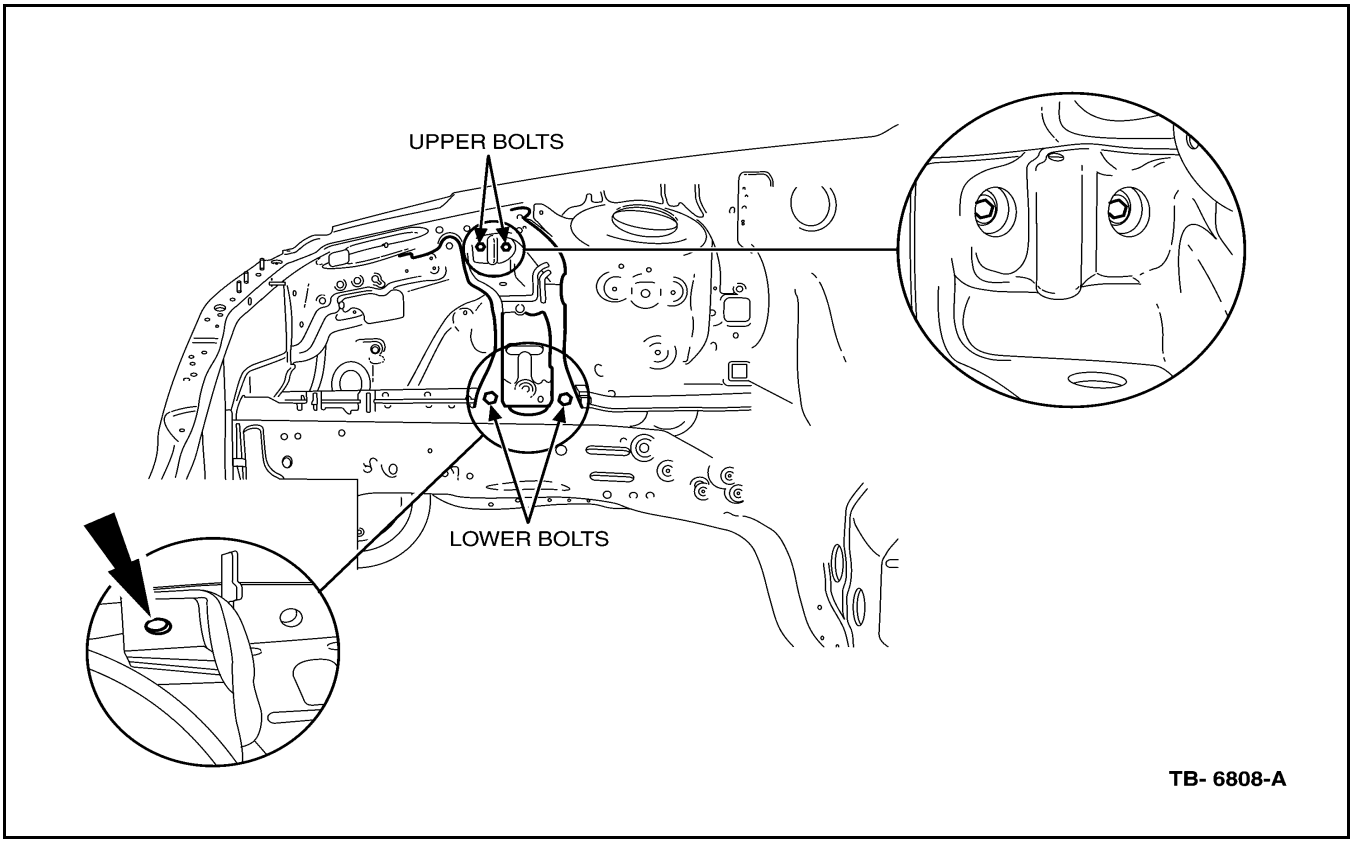


Figure 7 - Article 02-21-12

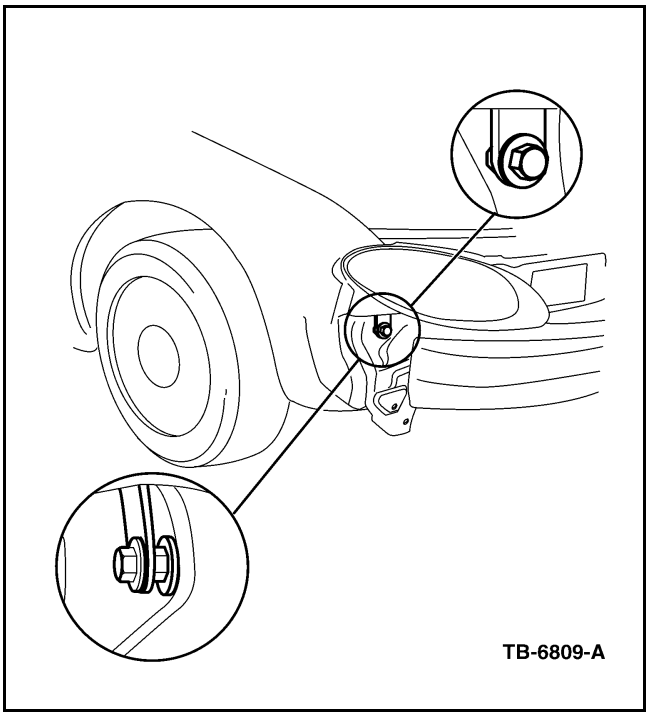


Figure 8 - Article 02-21-12

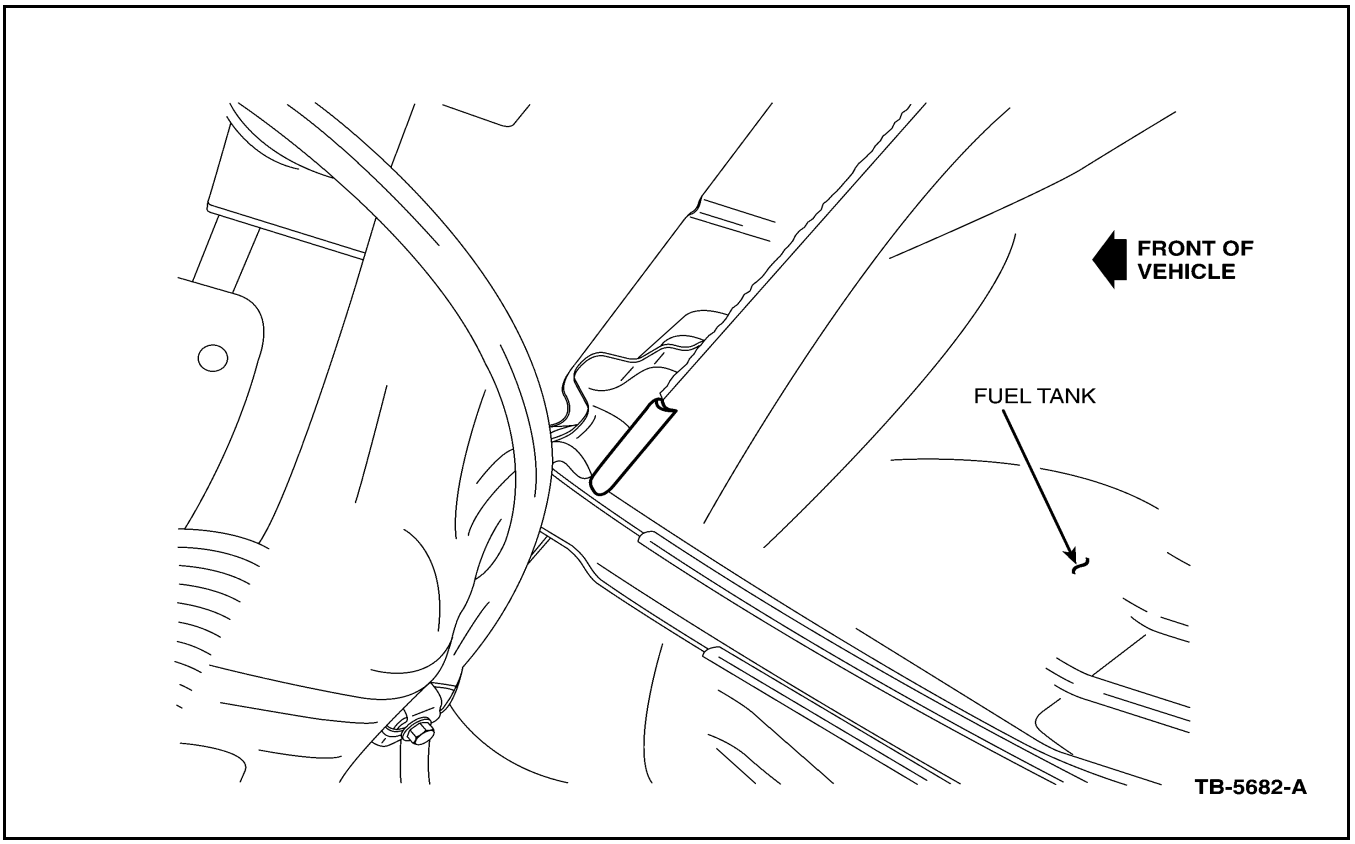


Figure 9 - Article 02-21-12

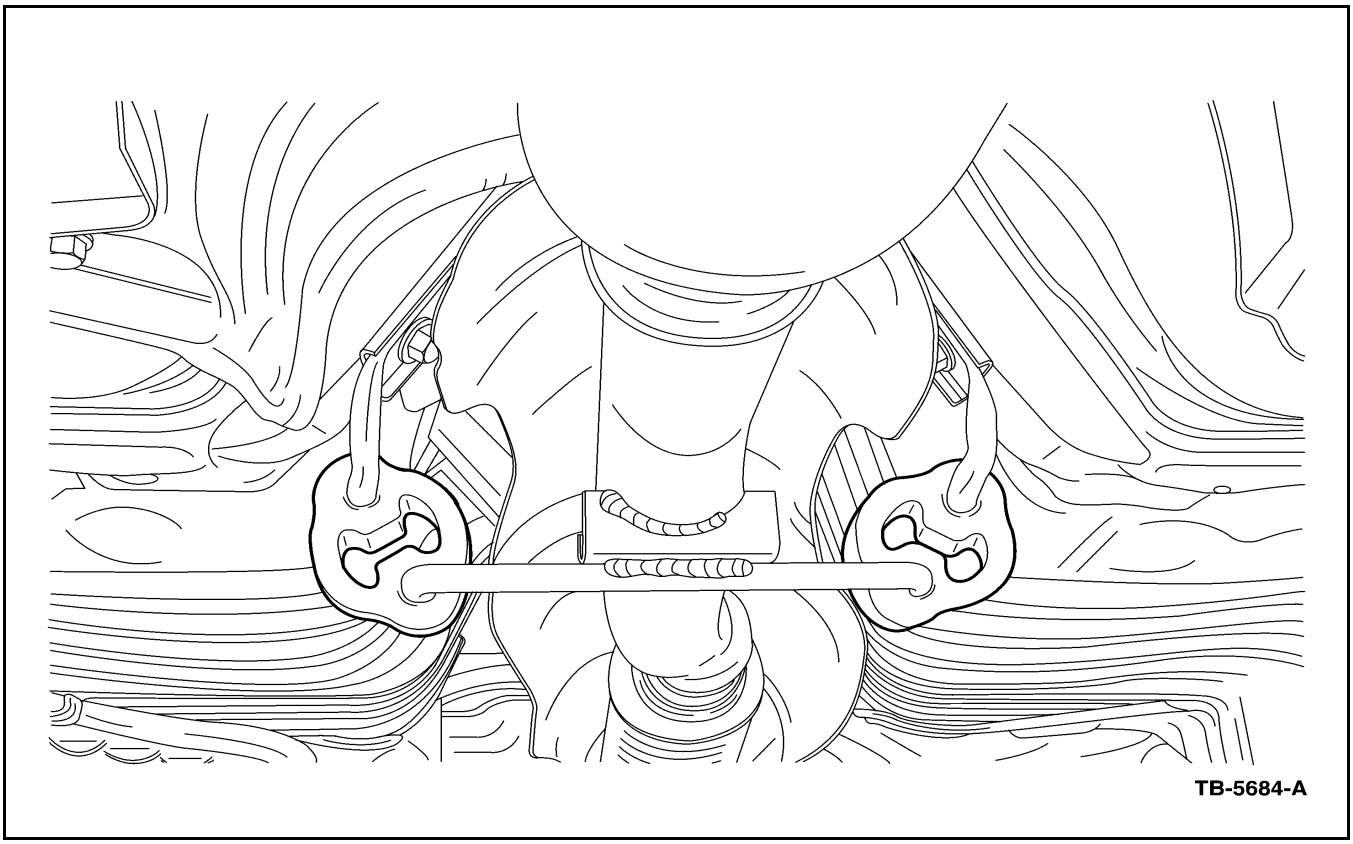


Figure 10 - Article 02-21-12

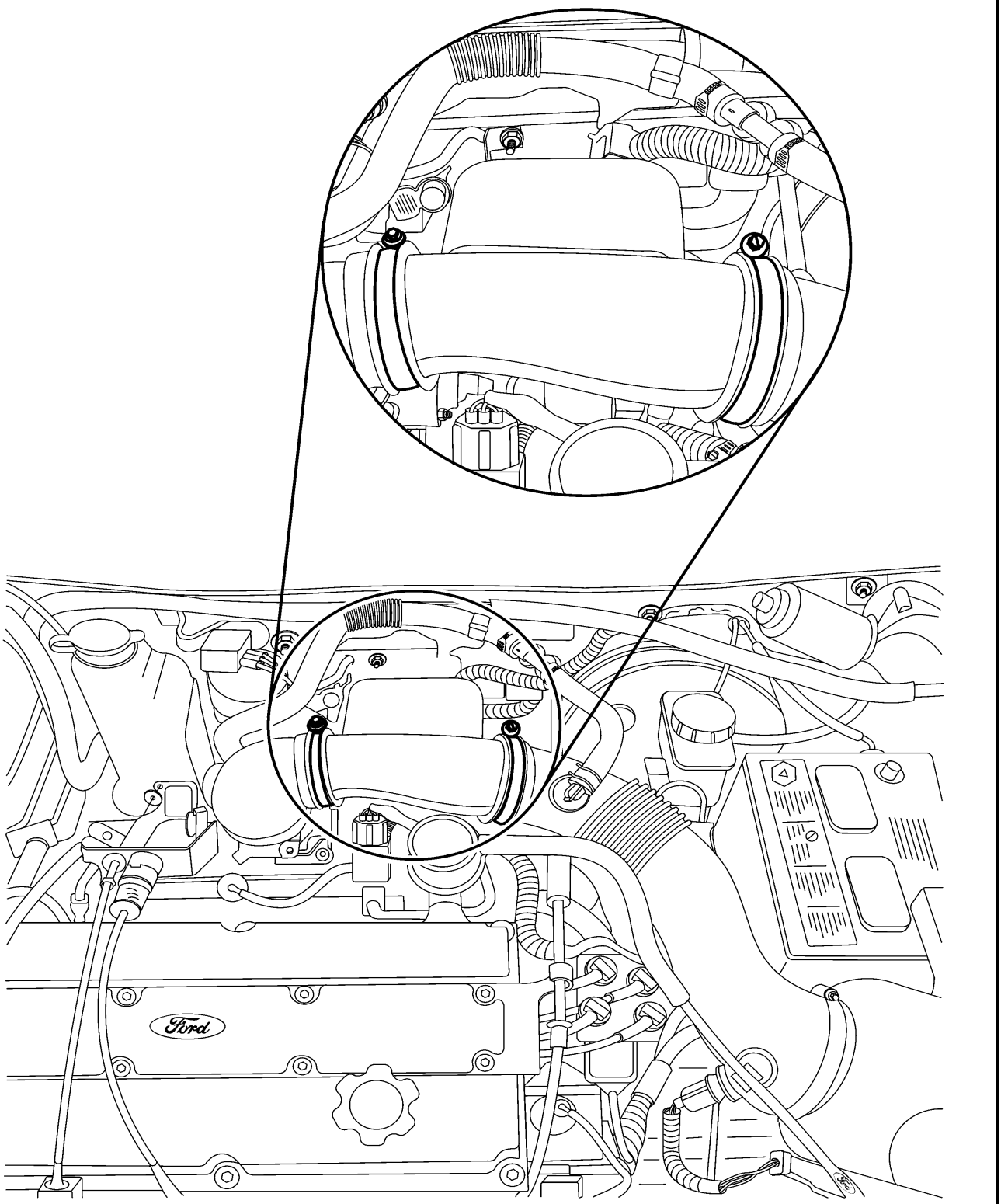


Figure 11 - Article 02-21-12

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by “do-it-yourselfers”. Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.